

Nickname:

12.5 Classwork: Roots of Complex Numbers (no calculator)

Answer all four problems in the spaces provided. Show your working. Give all answers as exact values (surds and fractions of π), not decimals.

1. (a) By writing $(a + bi)^2 = 5 - 12i$, where $a, b \in \mathbb{R}$, find the two square roots of $5 - 12i$.
(b) Hence solve the equation

$$z^2 - 2z - 4 + 12i = 0.$$

- (c) Explain why the two solutions in (b) are not a conjugate pair.

Nickname:

2. Let $w = -8 + 8\sqrt{3}i$.

- Write w in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.
- Solve the equation $z^4 = w$, giving each solution in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.
- Write the solution that lies in the first quadrant in the form $a + bi$.
- Hence write down the four solutions of $z^4 = -8 - 8\sqrt{3}i$, in the form $re^{i\theta}$.

Nickname:

3. (a) Find the three cube roots of 1, giving each in the form $a + bi$.
- (b) Hence solve the equation $(z - 1)^3 = 8$, giving each solution in the form $a + bi$.
- (c) Show that the equation in (b) can be written as $z^3 - 3z^2 + 3z - 9 = 0$. Hence write $z^3 - 3z^2 + 3z - 9$ as the product of a linear factor and a quadratic factor with real coefficients.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

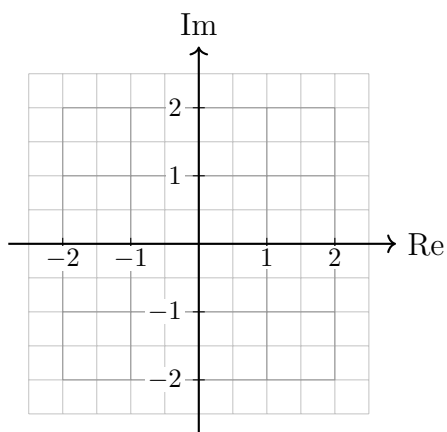
Nickname:

4. One of the cube roots of the complex number w is $1 + \sqrt{3}i$.

(a) Use de Moivre's theorem to find w .

(b) Find the other two cube roots of w , giving each in the form $a + bi$.

(c) The three cube roots of w are plotted on an Argand diagram. Find the exact area of the triangle they form.



.....

.....

.....

.....

.....

.....

.....

.....

.....